

KARTHIK JAYARAMAN

Staff SDET | Test Architect | QA Automation Leader

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PROFESSIONAL SUMMARY

Strategic Quality Leader with 23 years of experience in Software Quality Assurance and over 20 years architecting high-scale automation frameworks. Expert in C#, TypeScript/JavaScript, Java, and Python. Proven track record designing BDD-driven test ecosystems, leading CI/CD pipeline integration via Azure DevOps and CircleCI, and delivering measurable quality improvements across healthcare, fintech, and e-commerce domains.

- Architected and implemented comprehensive UI and API automation frameworks using **Playwright**, **Reqnroll** (Cucumber for .NET), RestSharp, and MSTest — serving as technical lead and governance owner for the automation ecosystem.
 - Authored and maintained **Azure Pipelines YAML** scripts to orchestrate multi-stage builds, schedule regression suite execution, publish test results and artifacts — enabling full CI/CD integration across development sprints.
 - Collaborated cross-functionally with Product and Engineering teams to define business-critical test strategies and robust test scenarios within Azure DevOps (ADO), ensuring maximum coverage of high-risk functional paths.
 - Engineered end-to-end performance and load testing strategy using JMeter + Azure Load Testing to facilitate microservices migration from on-premise to Azure Cloud — benchmarked to meet or exceed legacy performance standards.
 - Architected a high-performance test data generation utility using JMeter to simulate direct backend requests, **optimizing data seeding by over 90%** — enabling rapid generation of thousands of records without UI dependencies.
 - Proficient in mobile test automation (Appium + C#/TypeScript) for hybrid and native apps across iOS and Android.
 - 3+ years maintaining AWS Lambda functions in Python; experienced with ECS, ECR, EC2, DynamoDB, and SQS.
 - Strong background in ETL/data migration testing with complex SQL validation across SQL Server, Oracle, and DynamoDB.
 - Mentored and led QA teams of up to 10 engineers on test strategy, automation design, and framework best practices.
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TECHNICAL SKILLS

Languages	C#, TypeScript / JavaScript, Python, Java
UI Automation	Playwright (Reqnroll/MsTest), Selenium WebDriver, Cypress, WebdriverIO, Appium, Protractor
BDD Frameworks	Reqnroll (.NET), SpecFlow, Cucumber (Java & TS), Jasmine
API Testing	RestSharp (SpecFlow/MsTest), Postman, SoapUI
CI/CD & DevOps	Azure Pipelines (YAML), Azure DevOps Boards, CircleCI, GitHub Actions, TFS, GitHub
Perf & Load	JMeter, Azure Load Testing
Cloud Services	Azure Cloud, AWS (Lambda, ECS, ECR, EC2, SQS, DynamoDB), BrowserStack, LambdaTest, Sauce Labs
Databases	SQL Server, Oracle, DynamoDB, MongoDB
Tools	JIRA, Visual Studio, VS Code, ALM / HP Quality Center

WORK EXPERIENCE

Senior Test Automation Engineer | UNOS (United Network for Organ Sharing) | Richmond, VA | Jan 2024 – Present

Key Applications: UNet (Waitlist Management) | DonorNet (Donor & Match) | TransNet (Organ Shipment Tracking — Web & Mobile)

- Architected and implemented comprehensive automation frameworks for UI and API regression suites using **Playwright**, **Reqnroll** (Cucumber for .NET), RestSharp, and MSTest; served as technical mentor and governance owner.
- Authored **Azure Pipelines YAML** scripts to configure multi-stage builds, trigger automated regression runs on every code commit, and publish test results and code coverage artifacts to Azure DevOps dashboards.
- Collaborated with Product teams to define critical business workflows and high-risk areas; transformed requirements into comprehensive test strategies and scenarios within Azure DevOps (ADO).
- Engineered a performance and load testing strategy using **JMeter + Azure Load Testing** to benchmark microservices migration from on-premise to Azure Cloud, ensuring cloud environment met legacy performance standards.
- Architected a high-performance test data generation utility using JMeter, bypassing front-end dependencies by simulating direct backend requests — achieving **90%+ reduction in data seeding time**.
- Extended mobile automation coverage for the TransNet offline-capable mobile app using Appium and Appium Inspector with TypeScript.

Environment: Playwright, Reqnroll, Azure Pipelines (YAML), Azure DevOps, Azure Load Testing, JMeter, Appium, RestSharp, C#, SpecFlow, MsTest, Selenium WebDriver, Selenium Grid, SQL Server, Visual Studio

Staff Quality Engineer | Snagajob | Richmond, VA | Feb 2020 – Oct 2023

Online marketplace for hourly jobs connecting employers and job seekers across part-time and full-time roles.

- Built behavior-driven automation framework from scratch using Cucumber, TypeScript, and WebDriverIO with Page Object Pattern; maintained and enhanced with Protractor and Jasmine.
- Implemented CI/CD pipeline integration via CircleCI — automated UI and API regression suite execution on every code commit and published results to an internal regression reporting application.
- Implemented proof of concept comparing Cypress vs. WebDriverIO for e2e automation; provided recommendation that guided framework strategy.
- Created and maintained REST API automation scripts using C#, RestSharp, and SpecFlow; performed microservices testing across distributed services.
- Ran UI regression suite on AWS containers for cross-browser coverage; proficient in BrowserStack, Sauce Labs, and LambdaTest for cross-platform testing.
- Performed ETL/data migration testing with heavy SQL validation when migrating data from legacy RDBMS architecture to new architecture.
- Created scripts to sync test execution results from automation suite back to JIRA; maintained code in GitHub and contributed to epic/story backlog grooming.
- Mentored QA team members on test case design, automation methodology, and script development best practices.

Environment: WebDriverIO, Cypress, Protractor, TypeScript, Cucumber, C#, SpecFlow, RestSharp, Appium, CircleCI, AWS, GitHub, JIRA, VS Code, Angular

Senior Test Automation Engineer | UNOS (United Network for Organ Sharing) | Richmond, VA | Dec 2013 – Jan 2020

- Created and maintained BDD automation framework from scratch using SpecFlow, MSTest, C#, and Selenium WebDriver with Page Object Pattern.
- Created and maintained REST API automation scripts using C#, RestSharp, Entity Framework, and SpecFlow.

- Built test data generation tool using Microsoft Web Performance & Load Testing; ran Selenium Grid regression across IE, Chrome (Windows/Mac).
- Experienced in mobile test automation using Appium and C#; performed cross-browser and cross-platform test validation.
- Contributed to full-stack development: fixed C# defects, developed an MVC C# page in a patient safety project, and built Angular 2 components and a REST API for DonorNet Mobile.
- Performed data migration testing with complex SQL queries; created requirement traceability matrices and tracked defects through TFS.
- Mentored team members on test strategy, automation design, and Agile practices including backlog grooming and retrospectives.

Environment: Selenium WebDriver, Selenium Grid, SpecFlow, MSTest, C#, RestSharp, Entity Framework, Appium, TFS, SQL Server, Visual Studio, Angular 2, Page Object Pattern

QA Test Lead | Infosys | Multiple Locations, US | Sep 2003 – Dec 2013

Clients: Capital One Financial (Richmond, VA) | Ameriprise Financial (Minneapolis, MN) | ABN AMRO (Chicago, IL) | State Street Bank (India)

- Led QA for large-scale ETL and data migration initiatives at Capital One, validating billions of documents migrated between IBM Content Manager systems using Xenos Developer Studio.
- Automated regression test suites using HP QTP (keyword/data-driven/hybrid framework) for SIMS web application and SoapUI/Groovy for EI Web Service testing.
- Performed data warehousing and BI report validation (SAP Business Objects — Web Intelligence and Crystal Reports) across Oracle, MySQL, and DB2.
- Managed and mentored teams of up to 10 engineers; authored test plans, BAS documents, and requirement traceability matrices; coordinated offshore and client status calls.
- Developed Selenium WebDriver automation scripts with TestNG for web-based applications; applied UNIX scripting for backend data validation.
- Tested foreign exchange, fixed income, and online banking applications across Ameriprise, ABN AMRO, and State Street Bank engagements.

Environment: HP ALM, HP QTP, SoapUI, Selenium WebDriver, TestNG, SQL, Oracle, UNIX, SAP Business Objects, HP Quality Center 10.0, Java

CERTIFICATIONS

- Certified Software Tester (CSTE)
- Chartered Institute of Bankers in Scotland
- NSE's Certification in Financial Markets (NCFM)

EDUCATION

Bachelor of Engineering — Electrical and Electronics Engineering

Sri Sivasubramaniya Nadar College of Engineering (SSNCE), Madras University | Graduated 2003